

Aditya Bikram Thakur

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EDUCATION

Washington and Lee University

B.S. in Computer Science and Mathematics (double major); Full-ride scholarship (\$420K)

Lexington, VA

Expected May 2030

Budhanilkantha School

A Levels: Physics (A), Chemistry (A*), Computer Science (A*), Mathematics (A*)*

Valedictorian; SAT 1600

Kathmandu, Nepal

Aug. 2023 – May 2025

PROJECTS

DiatomMeter | Computer vision platform for SEM microscopy

Python, FastAPI, FastSAM, scikit-learn, OpenCV, React, TypeScript, Modal

diamometer.us | [GitHub](#)

Sep. 2026 – Present

- Building a platform with W&M's Nano & Biomaterials Lab that automates diatom analysis in SEM images.
- Designed a pipeline that calibrates scale, segments shells with FastSAM, filters debris, and grades damage.
- Measured 136 shells in one image in about 15 seconds on 2 CPUs, replacing one-at-a-time manual measurement.
- Trained a random forest on 44 multi-scale texture features to identify samples, abstaining when unsure.
- Reached 77% accuracy on held-out imaging sessions (86% when confident), beating 47% and 59% baselines.
- Validated output: scale within 0.12% of instrument calibration; median shell size inside published range.
- Deployed on Modal serverless behind a Cloudflare Worker, with a REST API, saved run history, and .xlsx export.

SATitude | Digital SAT prep platform

TypeScript, React 19, Vite, Express, Prisma, PostgreSQL, Clerk

satitude.xyz

Jul. 2026 – Aug. 2026

- Set up an npm-workspaces monorepo: a React 19 + Vite client and a TypeScript Express API hosted on Render.
- Synced Clerk accounts to PostgreSQL with idempotent Prisma upserts, fed by signed webhooks and a client hook.
- Validated request bodies with Zod and added Helmet headers, a CORS allow-list, and per-IP rate limiting.

walden.life | Interactive reading of Thoreau's *Walden*

JavaScript, React, Express, MongoDB, Mongoose

walden.life | [GitHub](#)

Jun. 2026 – Jul. 2026

- Built a full-stack app that maps *Walden's* 12 chapters onto 100 sourced daily entries on a timeline.
- Cut the timeline's initial payload 53% by serving a 5-field projection of each 10-field MongoDB record.
- Centralized fetching, loading, and error state in custom React hooks reused across every routed view.

EXPERIENCE

Incubate Nepal

Frontend Developer, [Nepalingo](#) (open source)

Kathmandu, Nepal

Jun. 2024 – Aug. 2024

- Merged 11 PRs, the 4th most of 10 contributors, into an open-source app teaching indigenous Nepali languages.
- Built the flashcard, daily quiz, and activity card components, with audio playback and loading/error states.
- Shipped in weekly sprints in React and TypeScript, with code reviewed by engineers from Vista and Cirkul.

Student Research Council Nepal

Backend Developer, [Lipi AI](#)

Kathmandu, Nepal

Jul. 2023 – Sep. 2023

- Built the backend of a mobile app that translates Tibetan inscriptions into English from a phone photo.
- Wrote segmentation for stacked Tibetan glyphs, which break the line-splitting assumptions of standard OCR.
- Adapted Tesseract OCR to Tibetan, a low-resource script, tuning preprocessing for low-contrast photos.
- Presented the working system to linguists as one of 6 students working under 2 mentors.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, HTML/CSS

Frameworks: React, Node.js, Express, FastAPI, Vite, Tailwind CSS, Three.js

ML & Vision: scikit-learn, OpenCV, scikit-image, NumPy, FastSAM, Tesseract OCR

Data & Cloud: PostgreSQL, Prisma, MongoDB, Docker, Git, Modal, Cloudflare Workers, Render

Certifications: [Stanford Algorithms](#) (4 courses), [Stanford & DeepLearning.AI Machine Learning](#), [Harvard CS50 AI](#)